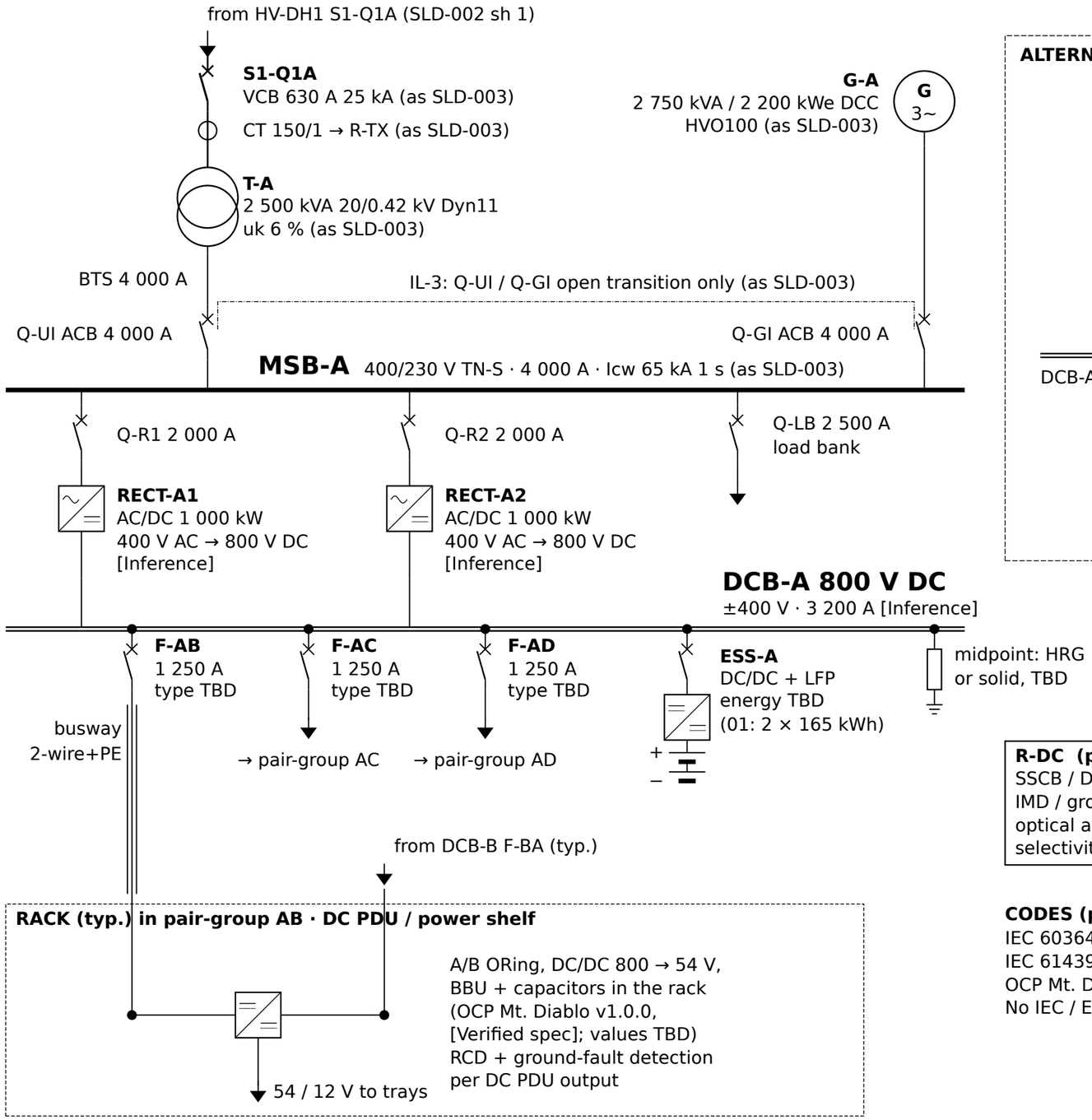
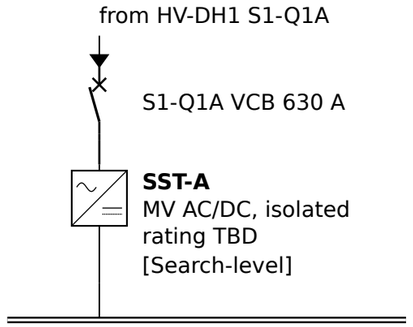


ALTERNATIVE B: SST 20 kV AC → 800 V DC (outline only)



Removes the T-A and RECT stages.
No LV point for G-A + IL-3: needs
MV gensets or an AC bus ahead
of the SST [Inference].
MV-side fault and arc study new.
OCP SST spec v0.3.0
[Search-level / Unverified]

R-DC (per DC feeder)
SSCB / DC trip: I>, di/dt (TBD)
IMD / ground fault (mandatory)
optical arc detection, DC rated (TBD)
selectivity with rack fuses (TBD)

CODES (pointers, [Unverified] unless opened)
IEC 60364 / SS 436 40 00 · EN 50600-2-2 · LVD 2014/35/EU
IEC 61439 · IEC 62040 · IEC 60947-2 (DC breakers)
OCP Mt. Diablo v1.0.0 [Verified] · NVIDIA 800 V DC [Vendor claim]
No IEC / EN 800 V DC PDU standard in sources read [Inference].

NOTES
1. Same block as SLD-003: 4N/3, 1.25 MW normal, 1.67 MW N-1 (83 % of 2 × 1 000 kW). MV, T-A, G-A, IL-3 kept.
2. Only the UPS / LV critical distribution is replaced. Aux, mechanical and AC loads stay as SLD-003.
3. Feeder current I = P / U: 0.83 MW / 800 V ≈ 1 040 A in N-1, 0.42 MW ≈ 520 A normal [Inference].
4. DC fault current = capacitor discharge + current-limited RECT / ESS. No IEEE 1584 method for DC; IEEE P3002.12 (DC arcing-fault guide) is an active PAR, not a standard [Verified, opened 2026-10-29].
5. Breaker type (SSCB / hybrid / DC MCCB), Icw, earthing, IMD and RCD settings: TBD from vendor data.
6. Labels: [Verified] = source opened · [Search-level] · [Inference] = own estimate. All ratings concept only.

v0.1 2026-10-29 Research draft for Project 02, not issued	
Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3 Portfolio D. Wu · Project 02 (draft) · public sources, scale NTS · A3	
Power block A — 800 V DC concept (draft) Alternative to SLD-003 · same 5 MW hall, 4N/3, blocks A–D CONCEPT / DRAFT — requires review by a licensed electrical engineer	
Drawing no. **DWU-DC-E-SLD-005**	Sheet **1 / 1** Rev **v0.1**